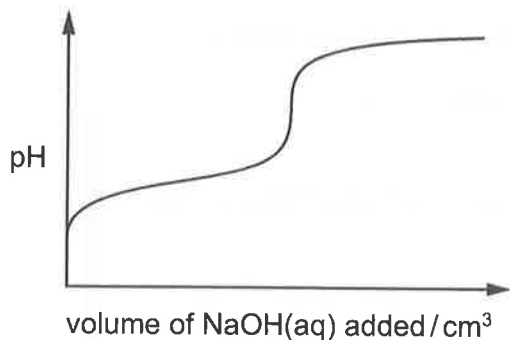


16 Use of the Data Booklet is relevant to this question.

In a titration experiment, 50.0 cm^3 of 0.100 mol dm^{-3} aqueous sodium hydroxide is slowly added to 25.0 cm^3 of 0.150 mol dm^{-3} aqueous ethanoic acid.

The pH of the reaction mixture is measured throughout the experiment and a graph of pH against volume of aqueous sodium hydroxide added is plotted.

Part of the titration curve is shown.



Consider the two statements.

- 1 The pH at the end of the experiment will be 13.0.
- 2 The pH when half of the acid has been neutralised will be the pK_a of the acid.

Which statements are correct?

- A** Both statements are correct.
- B** Neither statement is correct.
- C** Statement 1 only is correct.
- D** Statement 2 only is correct.